

**B.M.S. COLLEGE OF ENGINEERING, BANGALORE-19****Department of Computer Science and Engineering****INTERALS-1****Course Code: 23CS4PCADA****Course Title: Analysis and Design of Algorithms****Semester: 4****Maximum Marks: 40****Date: 08-04-2026****Faculty Handling the Course:**

Dr.KSS, PA, MS, Dr.JSN, Dr.BSR, Dr.NM, Dr.HSG, Dr.ST, AAS, DK, MNM, RP, HSM

Instructions: Internal choice provided in Part C.**PART-A****Total 5 Marks (No Choice)**

No.	Question	Marks	CO-PO	Bloom's Level
1	Solve the following recurrence relation using backward substitution method. i. $T(n) = T(n/2) + n$ for $n > 1$, $T(1) = 1$ 2.5M $T(n) = T(n/2) + n$ $= [T(n/2) + n/2] + n$ $= T(n/2) + n/2 + n$ $= T(n/2^3) + \frac{n}{2^2} + \frac{n}{2} + n$ $= T(n/2^k) + \frac{n}{2^{k-1}} + \frac{n}{2^{k-2}} + \dots + \frac{n}{2} + n$ $= T(n/2^k) + n \left[\frac{1}{2^{k-1}} + \frac{1}{2^{k-2}} + \dots + \frac{1}{2} + 1 \right]$ Assuming $n = 2^k$, then $k = \log_2 n$, we get $= T(1) + n [1 + 1]$ $= 1 + 2n$ $T(n) \in O(n)$	5M 1M 1M 0.5M	1-2	L4

	ii. $X(n)=2*X(n-1) + 1$ for $n>1$ and $X(0)=0$ 2.5M $ \begin{aligned} X(n) &= 2X(n-1) + 1 \\ &= 2[2X(n-2) + 1] + 1 \\ &= 2^2X(n-2) + 2 + 1 \\ &= 2^2[2X(n-3) + 1] + 2 + 1 \\ &= 2^3X(n-3) + 2^2 + 2 + 1 \\ &\vdots \\ &= 2^kX(n-k) + 2^{k-1} + 2^{k-2} + \dots + 2^2 + 2^1 + 1 \\ \text{Assuming } n-k &= 0, \text{ we get} \\ &= 2^kX(0) + \underbrace{1 + 2 + 2^2 + \dots + 2^{k-1}} \\ &= 2^k \cdot 0 + 2^k - 1 \\ &= 2^n \cdot 0 + 2^n - 1 \quad [\because n=k] \\ X(n) &\approx 2^n \quad [\because X(n) \in O(2^n)] \end{aligned} $	1M		
		1M		
		0.5M		

PART-B

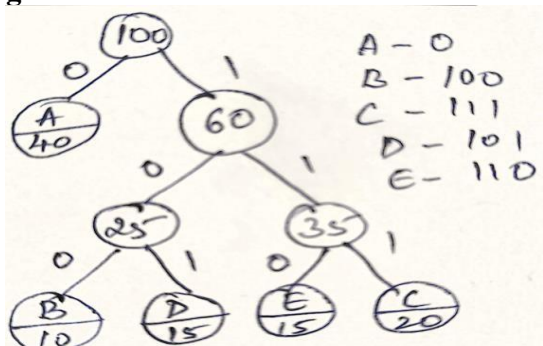
Total 15 Marks (No Choice)

No.	Question	Marks	CO-PO	Bloom's Level
2a)	A list contains employee records sorted by department. Now, it needs to be sorted by salary without disturbing department order for equal salaries. Suggest the best algorithm for this and justify. Also provide the algorithm and analyze its time efficiency. Merge Sort Identification: 1M Mergesort Algorithm: 1M Merge Algorithm: 2M Time Efficiency = 1M	5	1-2	L4
2b)	Consider the algorithm given below and answer the questions that follow: ALGORITHM Enigma ($A[0..n-1, 0..n-1]$) //Input: A matrix $A[0..n-1, 0..n-1]$ of real numbers for $i \leftarrow 0$ to $n-2$ do for $j \leftarrow i+1$ to $n-1$ do if $A[i, j] \neq A[j, i]$ return false return true i) What does this algorithm compute? The algorithm determines whether a square matrix is symmetric about its main diagonal. 1M ii) What is its basic operation? Comparison operation 1M iii) How many times is the basic operation executed? n^2 times 1M iv) What is the order of growth of this algorithm? $\theta(n^2)$ 2M	5	1-2	L4

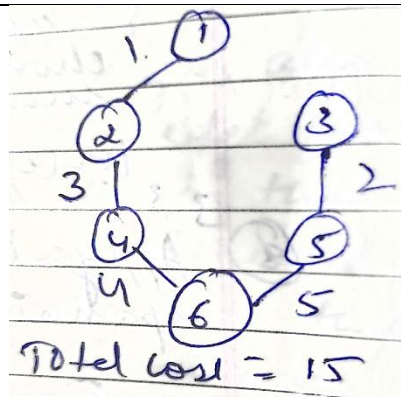
2c)	Apply Divide and Conquer technique to multiply the following two long integers: 1234 and 5678. Justify the significance of divide and conquer technique over the traditional method. Solution: Let $x=1234$, $y=5678$ Split it into two halves: $a=12$, $b=34$ $c=56$, $d=78$ Compute sub-products 2M $ac=12 \times 56=672$ $bd=34 \times 78=2652$ $ad=12 \times 78=936$ $bc=34 \times 56=1904$ $ad+bc=936+1904=2840$ Combine results 1M $xy = 672 \times 10^4 + 2840 \times 10^2 + 2652$ $= 6720000 + 284000 + 2652$ $= \mathbf{7006652}$ Justification: 2M The traditional method algorithm takes $O(n^2)$ The divide and conquer technique takes $O(n^{1.585})$ The number of multiplications is reduced to 3 using divide and conquer, which reduces the time taken by the algorithm.	5	2-1	L3
-----	--	---	-----	----

PART- C

Total 20 Marks

No.	Question	Marks	CO-PO	Bloom's Level																		
3a)	A university maintains a large database of student records, where certain attributes (like grades, attendance status, and course outcomes) are stored as symbols. Due to increasing data volume, the university plans to compress student data for efficient storage and faster retrieval. Design an efficient encoding algorithm to compress the student data. Given the following symbols and their frequencies: <table border="1"><thead><tr><th>Symbol</th><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th></tr></thead><tbody><tr><th>Description</th><td>Outstanding</td><td>Distinction</td><td>First Class</td><td>Second Class</td><td>Pass</td></tr><tr><th>Frequency</th><td>40</td><td>10</td><td>20</td><td>15</td><td>15</td></tr></tbody></table> i) Encode ABACABAD and ii) Decode 100010111001010 using the generated code. Identification of Huffman Code: 1M Problem Solving: 5M 	Symbol	A	B	C	D	E	Description	Outstanding	Distinction	First Class	Second Class	Pass	Frequency	40	10	20	15	15	10	2-1	L3
Symbol	A	B	C	D	E																	
Description	Outstanding	Distinction	First Class	Second Class	Pass																	
Frequency	40	10	20	15	15																	

Encoding ABACABAD: 2M Decoding 100010111001010: 2M				
OR				
3b)	A hospital needs to send an ambulance to reach patients across different locations. The travel time between locations due to road conditions and traffic are given below. Apply an appropriate algorithm to determine the shortest time required for the ambulance to reach all locations from the hospital. Also analyze the algorithm for its time efficiency. Identification of Dijkstra's: 2M Problem Solving with steps: 3M Final Shortest distances: 2M H to H = 0 H to 2 = 66 H to 3 = 17 H to 4 = 39 H to 5 = 52 H to 6 = ∞ Time efficiency: 3M	10	2-1	L3
4a)	Show how the following array is sorted using Quick Sort. Also, write the quick sort algorithm. 54, 26, 93, 17, 77, 31, 44, 55, 20 Quick Sort Algorithm = 2M Partition Algorithm = 3M Problem solving with steps = 4M 17, 20, 26, 31, 44, 54, 55, 77, 93 1M	10	2-1	L3
OR				
4b)	Find the MST for the following Graph using Kruskal's Algorithm and also write the Kruskal's Algorithm. Kruskal's Algorithm = 5M Problem solving with steps = 4M Total Cost = 1M	10	2-1	L3



ALL THE VERY BEST